

LLM-Based Agentic Monitoring vs. Classical Change-Point Detection for Financial Market Regime Identification

Oscar Pham & Paul Chen

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Abstract

We present a preregistered experiment comparing an LLM-based agentic monitoring system to classical change-point detectors for identifying regime breaks in two quantitative trading strategies. The agentic system integrates a small language model (qwen2.5:1.5b with grammar-constrained decoding) with filtered financial news, macroeconomic vintages, and strategy telemetry through a triage-escalation architecture; the classical arm uses Page-Hinkley, BOCPD, HMM, and distributional-threshold detectors. Both arms observe identical point-in-time data across 12 historical windows (2004–2020), split into a development set and a sequestered confirmatory test set. On the development set (8 event pairs), the agentic system detects regime breaks 9.75 days earlier on average (exact permutation $p = 0.0039$, Holm-adjusted $p = 0.0156$) with recall 8/8 versus 6/8 for classical. The test set (5 pairs) is directionally consistent (-4.4 days, $p = 0.25$) but structurally underpowered. The false-positive-rate co-primary endpoint does not favour the agentic arm. A three-condition leakage harness bounds pretraining memorisation, and an exploratory PnL simulation suggests the latency advantage translates to meaningful drawdown reduction. The results provide suggestive but not confirmatory evidence that multi-signal LLM fusion reduces detection latency relative to classical statistical methods.

1 Introduction

Quantitative trading strategies are subject to regime breaks. The August 2007 quant meltdown (Khandani & Lo, 2007), the 2009 momentum crash (Daniel & Moskowitz, 2016), and the February 2018 volatility event demonstrate that such breaks can destroy years of accumulated profit within days. Classical change-point methods, Page-Hinkley (1954), Bayesian online change-point detection (Adams & MacKay, 2007), hidden Markov models (Hamilton, 1989), and threshold rules, operate exclusively on the strategy’s return stream and react only after its statistical properties have already shifted. If news coverage of a developing crisis precedes its statistical manifestation in returns, an LLM-based monitor that fuses telemetry with natural-language context might detect regime breaks earlier.

We report a preregistered experiment testing three hypotheses: **H1 (co-primary)**: the agentic system achieves both lower detection latency and lower false-positive rate (FPR) than the best classical detector; **H2**: any advantage generalises across both strategies; **H3**: the magnitude of advantage differs by strategy.

Our contributions are: (1) a three-layer agentic monitoring architecture running a 1.5B-parameter model on consumer hardware; (2) an evaluation protocol with strict information parity on a survivorship-bias-free universe, a distinction that reverses the sign of the anchor event ($+3.4\%$ on a survivor-only universe against a -4.8% drawdown point-in-time); (3) a three-condition leakage harness separating genuine detection from recall of famous events; and (4) full disclosure of protocol deviations and operational failure modes.

2 Data and Strategies

All inputs are constructed point-in-time (PIT): on any decision date t , both arms observe only values published on or before t . The eligible universe is reconstructed from historical S&P 500 constituent membership intervals (~ 347 eligible names per month, versus ~ 542 under a survivor-only universe). Two daily PnL series are built on this universe:

AL_PCA (Avellaneda & Lee, 2010): sector-sleeve PCA mean reversion ($K = 15$ components; Ornstein–Uhlenbeck residual z-scores with fixed entry/exit thresholds), spanning 2003–2014. **JT_MOM** (Jegadeesh & Titman, 1993): cross-sectional momentum (12-1 ranking, long top decile, short bottom decile, monthly rebalance), spanning 2003–2020. Daily return correlation between the two series is ~ 0.003 ; this near-orthogonality is what makes the cross-strategy hypotheses informative. The AL_PCA curve ends in 2014, excluding it from three of four test-set events and reducing the confirmatory sample to five pairs (Section 5).

News context comes from the FNSPID corpus (Dong et al., 2024): $\sim 11.8\text{M}$ articles (2003–2020) in per-year Parquet files, queried over a 7-day trailing window with a risk-term filter. Article density is uneven ($\sim 2,800/\text{day}$ in 2007–2016, $\sim 80/\text{day}$ pre-2007, $\sim 1,400/\text{day}$ 2017–2020); the NYT Archive API supplements sparse pre-2010 windows. Macroeconomic context uses ALFRED *vintage* series (first-print values as published on the decision date) rather than revised FRED series, closing a revision-lookahead channel.

3 Method

3.1 Classical Arm

Four detectors operate on the daily return stream: Page-Hinkley ($\delta = 0.5$, $\lambda = 8.0$), BOCPD (hazard 1/250), a two-state Gaussian HMM (low- to high-volatility transition), and a rolling z-score threshold. An aggregate alarm fires when ≥ 2 detectors fire within 5 trading days. For the primary H1 analysis the best single detector (HMM) is the headline comparator, since the aggregation rule’s latency penalty makes it the weaker (less conservative) baseline.

3.2 Agentic Arm

Each trading day passes through three layers (Figure 1): *triage*, a *news context agent*, and a *performance supervisor*, sharing a guardrail module that enforces information parity with the classical arm.

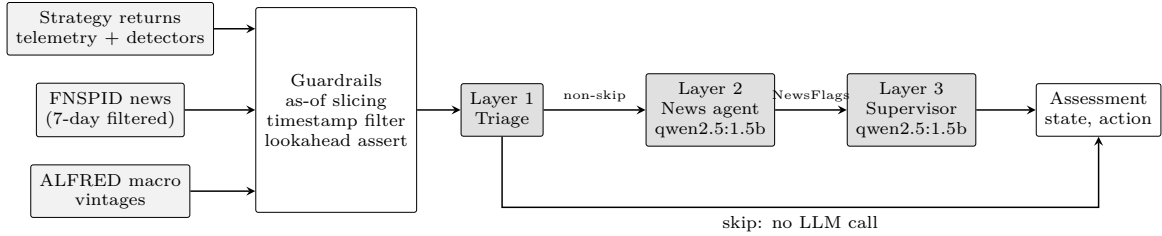


Figure 1: Agentic architecture. All inputs pass through fail-closed guardrails before the three-layer pipeline; triage routes each day to SKIP (no LLM call) or to progressively richer inference modes.

Triage. Three causal signals set the per-day compute budget: FinBERT stress s (daily-maximum negative-sentiment probability over filtered headlines), a trailing-baseline news-intensity z-score of risk-term hits, and the count n of classical detectors alarming within 3 calendar days. The rule (fixed a priori) is

$$\text{mode} = \begin{cases} \text{CLASSICAL_ESCALATION} & \text{if } \geq 2 \text{ detectors within 5 days} \\ \text{THINKING} & \text{if } s \geq 0.60 \vee z \geq 2.0 \vee n \geq 1 \\ \text{CHEAP} & \text{if } s \geq 0.30 \\ \text{SKIP} & \text{otherwise.} \end{cases}$$

News context agent. Non-skip days send up to 40 risk-filtered articles from the 7-day trailing query to the model, which emits structured **NewsFlags** JSON (overall risk level, flagged evidence, narrative, confidence). Output is grammar-constrained (Ollama `format=json`); one repair retry is attempted, and persistent failures are logged and the day proceeds without news flags.

Performance supervisor. The supervisor receives causally sliced telemetry (trailing 60-day statistics, drawdown), classical detector alarms, ALFRED macro context, and the news flags, and emits an assessment with `state` $\in \{\text{NORMAL, WATCH, ALERT, CRITICAL}\}$ and an action recommendation. A day is an *alarm* if `state` $\in \{\text{ALERT, CRITICAL}\}$; alarm clusters are de-duplicated with a 5-trading-day cooldown.

Guardrails. Four fail-closed checks enforce parity: (1) as-of slicing of telemetry and alarms; (2) a publication-timestamp filter on articles; (3) a future-text scrub dropping articles embedding any ISO date past the decision date; (4) a recursive lookahead assertion over the full context that raises a hard error on violation.

Model. Both agents use qwen2.5:1.5b via Ollama at temperature 0. The preregistered llama3.2:3b was substituted before any evaluation after failing structured-output benchmarks (11/30 valid JSON vs. 30/30; Deviation 1). Full prompts and schemas are in the repository.

3.3 Endpoints and Analysis Plan

All metrics, tests, and decision rules were fixed before an evaluation freeze (git tag `eval-freeze-v1`); no thresholds were tuned on evaluation data.

Latency (co-primary). For each event window with preregistered onset t_{onset} , latency is $\ell = t_{\text{alarm}} - t_{\text{onset}}$ (calendar days) for the first alarm in $[t_{\text{onset}}, t_{\text{onset}} + 21\text{d}]$ (agentic: ALERT/CRITICAL; classical: HMM). Misses are right-censored at $\ell = 21$ so non-detections enter as worst-case latencies.

FPR (co-primary). During calm windows every alarm is false. Agentic FPR counts de-duplicated cluster starts per trading day; classical FPR counts every alarm day. This asymmetry (Deviation 9) biases in favour of the agentic arm, so a null FPR result is robust a fortiori.

Secondary. Recall (binary detection in the 21-day zone; runtime failures count as misses) and an onset-sensitivity re-scoring against curve-derived onsets (peak before deepest trough).

Tests. Latency: exact permutation over all 2^n sign-flips of paired differences (one-sided). FPR: moving-block bootstrap ($b = 10\text{d}$, $B = 10,000$, seed 42; two-sided), with Fisher’s exact as a check. Recall: McNemar exact and a Beta–Binomial posterior (Jeffreys prior). The two co-primary p -values are Holm-corrected ($k = 2$); H1 is *supported* if both reject at family-wise $\alpha = 0.05$, *partially supported* if one rejects. H2 is assessed directionally per strategy; H3 as the interaction $\Delta_{\text{AL_PCA}} - \Delta_{\text{JT_MOM}}$.

A priori power bound. With n pairs the permutation test’s minimum p is 2^{-n} : $1/256$ for the development set ($n = 8$) but $1/32$ for the test set ($n = 5$), whose Holm-adjusted best case $2/32 = 0.0625$ cannot reject at $\alpha = 0.05$. The confirmatory set is structurally underpowered, a limitation acknowledged before unblinding.

3.4 Evaluation Protocol

Twelve windows are split into a development set (events: quant meltdown 2007, GFC/Lehman 2008, momentum crash 2009, US downgrade 2011; calm: 2004–2006, 2013–2014) and a sequestered test set (events: flash crash 2010, China devaluation 2015, volmageddon 2018, COVID-19 2020; calm: 2012, 2017). The partition was fixed before any runs; test windows were accessed only after an automated freeze gate (11 checks) passed on the tagged codebase. After the tag, no prompt, threshold, or scoring rule changed; post-freeze infrastructure fixes (context caching, dead-runner recovery, thermal-stall reruns) affect availability only and are logged as Deviations 12–15.

To bound pretraining memorisation, a leakage harness runs three conditions: **A** (standard), **B** (all ISO dates replaced by a non-parseable sentinel, preserving numerical evidence while blocking memorised event–date associations), and **C** ($M = 10$ synthetic windows with fabricated news and shuffled returns). The preregistered bound is memorisation $\leq \text{perf}_A - \min(\text{perf}_B, \text{perf}_C)$, with detection classified evidence-driven if $\min(\text{perf}_B, \text{perf}_C) \geq 0.6 \text{perf}_A$.

All experiments ran on one consumer laptop (i7-11800H, 32 GB RAM, RTX 3050 Ti 4 GB VRAM; FinBERT on CPU to avoid VRAM contention). Fifteen deviations from the preregistration are fully documented in `docs/DEVIATIONS.md` in the repository; the most consequential are cited inline.

4 Results

4.1 Detection Latency

Table 1 reports all paired latencies. On the development set the agentic system detects 9.75 days earlier on average (exact permutation $p = 0.0039$; Holm $p = 0.0156$), rejecting H_0 . The test set is directionally consistent (-4.4 days) but not significant ($p = 0.25$, $n = 5$).

4.2 False-Positive Rate

Table 2 reports per-calm-window FPR. Development-set agentic FPR (1.31%) is statistically indistinguishable from classical HMM (1.08%): bootstrap $p = 0.62$ (95% CI for the difference $[-0.42, +0.88]$ pp), Holm $p = 0.62$. On the test set the agentic arm is *worse* (6.93% vs. 1.41%, $p = 0.51$), driven by calm 2012. There is no evidence of an agentic FPR advantage.

4.3 Recall and H1 Verdict

Development recall is 8/8 agentic vs. 6/8 classical (McNemar $p = 0.50$; posterior $P(\theta_A > \theta_C) = 0.936$); test recall is 4/5 vs. 3/5 (McNemar $p = 1.0$; posterior 0.75). Applying the preregistered rule, H1 is **partially supported**: latency rejects on the development set (Holm $p = 0.0156$) but FPR does not (Holm $p = 0.62$). On the test set neither endpoint rejects; the confirmatory analysis is inconclusive for lack of power.

Table 1: Paired detection latency (calendar days from onset). “Cens.” marks non-detections imputed at $\ell = 21$ d. Classical comparator is HMM.

Window	Strategy	Classical	Agentic	Δ	Set
Quant meltdown 2007	AL_PCA	7	0	−7	Dev
GFC / Lehman 2008	AL_PCA	14	0	−14	Dev
Momentum crash 2009	AL_PCA	1	0	−1	Dev
Downgrade 2011	AL_PCA	11	5	−6	Dev
Quant meltdown 2007	JT_MOM	7	0	−7	Dev
GFC / Lehman 2008	JT_MOM	Cens.	2	−19	Dev
Momentum crash 2009	JT_MOM	Cens.	1	−20	Dev
Downgrade 2011	JT_MOM	7	3	−4	Dev
Flash crash 2010	AL_PCA	8	5	−3	Test
Flash crash 2010	JT_MOM	Cens.	1	−20	Test
China deval. 2015	JT_MOM	2	9	+7	Test
Volmageddon 2018	JT_MOM	Cens.	8	−13	Test
COVID-19 2020	JT_MOM	14	Cens.	+7	Test
<i>Dev mean (8 pairs)</i>		11.1	1.38	−9.75	
<i>Test mean (5 pairs)</i>		13.2	8.8	−4.4	

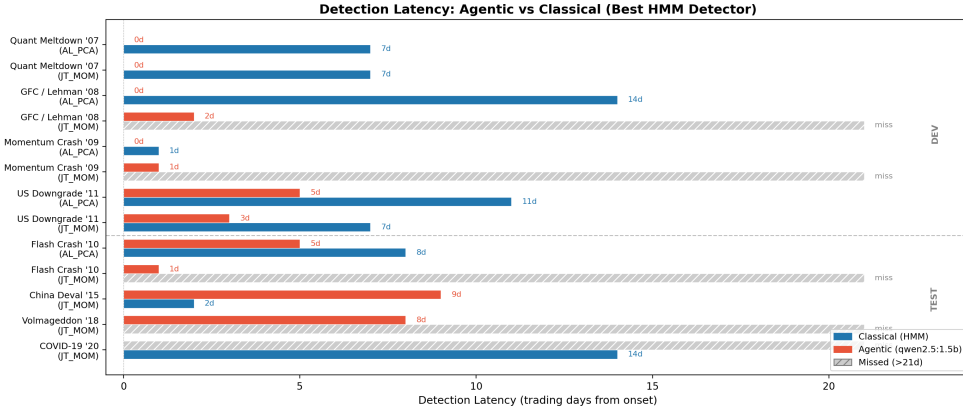


Figure 2: Detection latency per event window: agentic (red) versus classical HMM (blue); shorter bars are better. Hatched bars denote misses censored at 21 days.

Table 2: False-positive rate per calm window (agentic: alarm cluster starts / trading days; classical: alarm days / trading days, cf. Deviation 9).

Calm window	Strategy	Classical	Agentic	Set
2004–2006	AL_PCA	3.05%	0.13%	Dev
2004–2006	JT_MOM	0.53%	0.13%	Dev
2013–2014	AL_PCA	0.00%	3.84%	Dev
2013–2014	JT_MOM	0.20%	2.30%	Dev
2012	AL_PCA	1.20%	10.38%	Test
2012	JT_MOM	2.00%	7.69%	Test
2017	JT_MOM	1.20%	2.70%	Test
<i>Dev aggregate</i>		1.08%	1.31%	
<i>Test aggregate</i>		1.41%	6.93%	

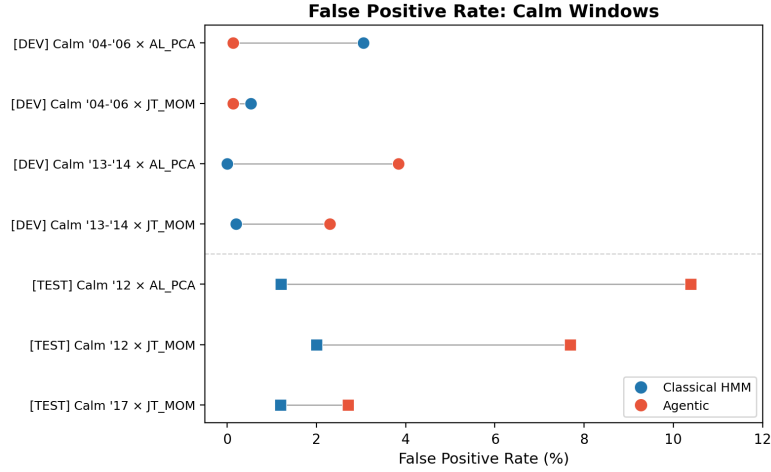


Figure 3: False-positive rate per calm window. Agentic FPR is comparable on the development set but elevated on the test set, particularly calm 2012.

4.4 Strategy Breakdown (H2, H3)

The latency advantage is directionally consistent across both strategies on the development set: AL_PCA mean $\Delta = -7.0$ d, JT_MOM -12.5 d (both 4 pairs), supporting H2 directionally; on the test set both directions remain agentic-favourable (AL_PCA -3 d on 1 pair; JT_MOM -4.75 d on 4). The H3 interaction is $+5.5$ d: JT_MOM benefits more (Figure 4), because the classical HMM missed GFC 2008 and momentum crash 2009 entirely on momentum returns while news coverage clearly signalled stress.

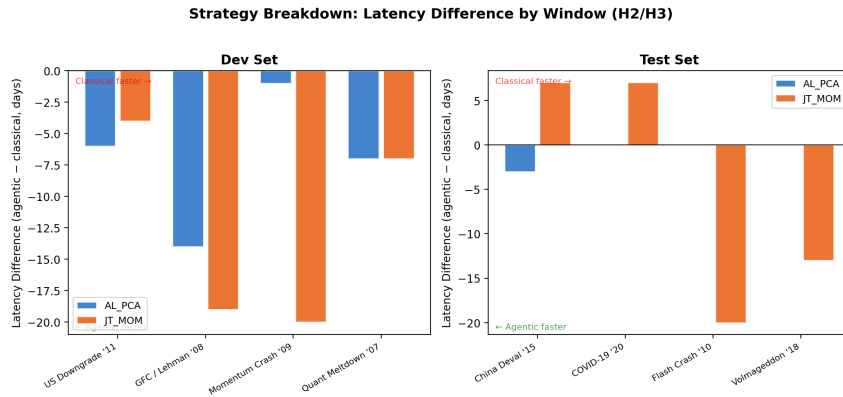


Figure 4: Per-strategy latency advantage (development set). JT_MOM benefits more, primarily because the HMM misses two events on momentum returns.

4.5 Economic Significance (Exploratory)

We simulate three regimes per event window: unmonitored, classical-monitored (flatten at first HMM alarm), and agentic-monitored (flatten at first ALERT or CRITICAL), with no transaction costs and no re-entry. Across the 13 event pairs (Table 3, Figure 5), mean maximum drawdown avoided is 1.82 pp (classical) versus 10.09 pp (agentic), an advantage of 8.28 pp per event. The extreme case is momentum crash 2009 on JT_MOM: the unmonitored strategy lost 94.47%, the HMM never fired, and the agentic exit avoided 72.28 pp. Excluding this window the mean advantage is 2.95 pp. China devaluation 2015 is the only reversal (-1.72 pp): the agentic alarm fired 21 July, exiting before a partial recovery, while the classical alarm on 13 August was better timed. These idealised assumptions overstate absolute value but affect both arms symmetrically.

4.6 Robustness, Leakage, and Operations

Onset sensitivity. Re-scoring against curve-derived onsets yields -16.4 d ($p = 0.0039$) on the development set and -11.8 d ($p = 0.0625$) on the test set; rank ordering is unchanged. Using the aggregate rule instead of

Table 3: Maximum drawdown avoided (pp) relative to the unmonitored baseline.

Window	Strategy	Classical	Agentic	Advantage
Quant meltdown 2007	AL_PCA	3.05	6.90	3.85
GFC / Lehman 2008	AL_PCA	1.82	3.16	1.34
Momentum crash 2009	AL_PCA	2.10	4.00	1.90
Downgrade 2011	AL_PCA	0.00	0.06	0.06
Quant meltdown 2007	JT_MOM	0.00	1.47	1.47
GFC / Lehman 2008	JT_MOM	0.00	15.63	15.63
Momentum crash 2009	JT_MOM	0.00	72.28	72.28
Downgrade 2011	JT_MOM	4.66	7.30	2.64
Flash crash 2010	AL_PCA	2.38	3.49	1.11
Flash crash 2010	JT_MOM	6.15	10.34	4.19
China deval. 2015	JT_MOM	3.44	1.72	-1.72
Volmageddon 2018	JT_MOM	0.00	0.00	0.00
COVID-19 2020	JT_MOM	0.00	4.87	4.87
<i>Mean</i>		1.82	10.09	8.28

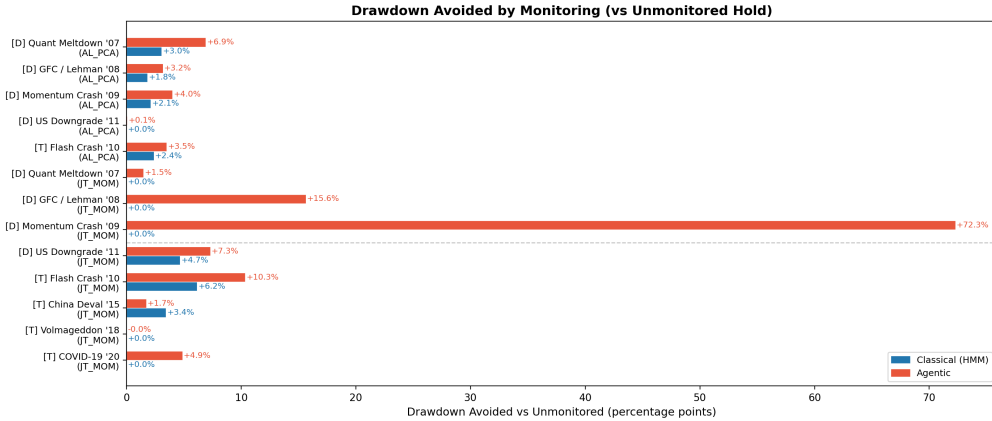


Figure 5: Drawdown avoided (pp) per event window. The agentic arm avoids more drawdown in 11 of 13 windows (one tie, one reversal).

the HMM as comparator widens the agentic advantage (aggregate detected 4/8 dev events vs. HMM’s 6/8), so the HMM is the conservative baseline.

Leakage bound. On flash crash 2010 (AL_PCA), Condition A detected with latency 5 d, Condition B (date-masked) with latency 0 d, and Condition C recalled 10/10 synthetic windows. With $\text{perf}_A = \text{perf}_B = \text{perf}_C = 1.0$, the memorisation bound is ≤ 0 and detection is classified **evidence-driven**. The $B \approx A$ result directly manipulates the date channel; the high Condition C recall should be read cautiously since synthetic windows use an untrained HMM.

Operational metrics. Mean LLM calls per trading day are 1.09 (0.10 on sparse-news calm 2004–2006 to 2.0 on quant meltdown 2007); wall-clock cost is 4.6 s typical and 8.4 s worst case per day, so a full window completes in 6–12 minutes, feasible for end-of-day monitoring. News-agent JSON failures on non-skip days range from 0% to 79% (dense-news calm windows): failures suppress alarms, so they bias *against* the agentic arm for latency but *in favour* of it for FPR (Deviation 7); the latency advantage is therefore conservative, the FPR comparison is not.

5 Discussion

Why earlier. Two mechanisms drive the latency advantage. First, news anticipation: coverage of developing crises precedes their statistical manifestation in returns (quant-meltdown coverage of crowded quantitative strategies appeared in late July 2007, enabling 0-day latency where the HMM needed 7 days of anomalous returns). Second, multi-signal fusion: the supervisor can escalate on the conjunction of individually sub-threshold signals, whereas classical detectors observe returns alone. The advantage is largest where the return stream never exhibits the volatility signature the HMM is calibrated for (GFC 2008 and momentum crash 2009 on JT_MOM).

Why FPR is not better. FinBERT stress saturates near 0.97 on essentially every day with news (mean 0.971 on calm 2013–2014 and on GFC 2008 alike), so the 0.60 triage threshold degenerates into a “does news

exist today?” gate: dense-news days bypass skip mode, and the 1.5B supervisor lacks the precision to separate concerning signals from routine financial noise, over-escalating on calm windows.

Power and the COVID-19 miss. The 5-pair test set cannot reject after Holm correction even in the best case (Section 3.3); this follows from the AL_PCA curve ending in 2014, not from the results. The agentic arm’s only test-set miss is COVID-19 2020. News was available (40 articles/day across all 82 days), but early pandemic coverage was dominated by ambiguous general-volatility headlines; the classical HMM detected at 14 d from returns alone. A 1.5B model may lack the capacity to interpret an unprecedented shock as a regime change from ambiguous text.

Limitations. (1) Small samples: 8 development and 5 test pairs; the confirmatory set cannot reach significance. (2) Single model and single news corpus; findings may not transfer. (3) The FPR scoring asymmetry favours the agentic arm, making the null FPR result conservative but not symmetric. (4) The PnL simulation is frictionless with no re-entry. (5) Famous, heavily documented test events carry memorisation risk; the leakage harness covers one test window. (6) US large-cap equities only. (7) Fifteen protocol deviations, mostly infrastructure, are disclosed in the repository; none alter prompts, thresholds, or scoring post-freeze.

6 Conclusion

In this preregistered comparison, an LLM-based agentic monitor detected regime breaks significantly earlier than classical change-point detection on the development set (-9.75 d, Holm $p = 0.0156$; recall 8/8 vs. 6/8), with a directionally consistent but underpowered test set (-4.4 d, $p = 0.25$). The FPR co-primary endpoint did not favour the agentic arm, tracing to saturation of the sentiment-based triage gate. Multi-signal fusion of returns, news, and macro context thus shows genuine promise for earlier detection, with exploratory economic value (8.28 pp mean drawdown avoided per event), but false-positive control requires redesigned triage (change-in-sentiment or learned routing), larger models for supervisor precision, longer backtest curves for confirmatory power, and leakage harness coverage of all test windows.

Data and Code Availability

Code, prompts, result artifacts, and the full deviation log are available at <https://github.com/PC1101/In-Market-Agentic-AI-Monitoring-Application-in-Financial-Market> (tag `eval-freeze-v1` pins the evaluated code). The FNSPID corpus is subject to its original license and must be obtained from the source; all derived evaluation artifacts are included. Experiments used Ollama (qwen2.5:1.5b) and ProsusAI/FinBERT on consumer hardware.

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A Protocol Deviations

Table 4 summarises all 15 deviations from the preregistered protocol; full narratives with evidence and timestamps are maintained in the repository (`docs/DEVIATIONS.md`).

Table 4: Protocol deviations. “Bias” indicates whether the deviation favours or disfavours the agentic arm for the affected endpoint.

#	Preregistered	Actual	Rationale	Bias direction
1	llama3.2:3b	qwen2.5:1.5b	Failed JSON benchmarks (11/30 vs. 30/30); substituted before eval	Smaller model
2	90% completeness	80% completeness	Include gfc_2008 AL (84%); all missing days post-detection	Includes 1 more pair
3	AL_PCA on china_deval	Excluded	PIT curve ends 2014	Fewer test pairs
4	Triage skips calm days	0% skip on 2013–2014	FinBERT saturation	Inflates FPR (against agentic) Includes gfc JT
5	Supervisor-day completeness	Triage-day completeness	Triage = attempted; supervisor = succeeded	None
6	Stub latencies	Real-model latencies	Stub was provisional	Latency: against; FPR: for agentic
7	Not specified	News-agent failures documented	0–79% failure rate on non-skip days	Flatters agentic latency
8	Calibrated detector params	Default params	Calibrated grid computed but never wired in; HMM unaffected	Favours agentic FPR
9	Symmetric FPR scoring	Asymmetric (cluster vs. alarm-day)	Analysis code difference	Partial coverage
10	Leakage on all test events	One test window covered	Harness incomplete at freeze	None
11	Single test-set pass	Rerun after runner death	Infrastructure outage, not result-shopping	Availability only
12	No resume logic	Day-level resume + recovery	Availability fix; identical prompts at $T = 0$	None (identical inputs)
13	Inline news queries	Two-pass context cache	VRAM contention on covid_2020	Availability only
14	No per-day retry	15s cooldown + ping + 1 retry	GPU thermal throttle	Conservative (less info)
15	Full 40-article cap	10 articles on 3 thermal-stall days	Prevent re-triggering stall	

B Triage Hyperparameters

Table 5: Triage hyperparameters (fixed a priori, preregistration §6.2).

Parameter	Value	Purpose
STRESS_CHEAP	0.30	FinBERT stress $\rightarrow$ cheap mode
STRESS_THINKING	0.60	FinBERT stress $\rightarrow$ thinking mode
HIT_Z_THINKING	2.0	News intensity $z \rightarrow$ thinking mode
RECENT_DAYS	3	Detector lookback (calendar days)
COVERAGE_DAYS	7	News context window (calendar days)
BASELINE_PAD	120	z -score baseline history (calendar days)
Temperature	0.0	Greedy decoding (deterministic)
Max articles	40	Per-day article cap to news agent